

SONOMA COUNTY AP, CA, USA

WMO: 724957

Lat: 38.504N

Lon: 122.810W

Elev: 35

StdP: 100.91

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23213

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.5	-0.3	-7.3	2.0	14.0	-4.3	2.6	10.6	9.4	12.9	8.1	12.4	0.7	300	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	34.9	19.2	32.8	18.7	30.7	18.1	20.6	32.2	19.7	30.7	18.8	29.1	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	11.5	23.7	15.1	10.8	22.6	14.4	10.3	21.7	59.1	32.5	55.8	30.8	53.2	29.1	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.7	5.7	DB	-3.7	38.7	1.3	1.8	-4.6	40.0	-5.4	41.1	-6.1	42.1	-7.0	43.4	(4)
(5)			WB	-4.2	22.7	1.4	2.0	-5.2	24.2	-6.0	25.3	-6.8	26.5	-7.8	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	8.8	10.2	11.8	13.3	16.0	18.8	19.7	19.8	19.0	16.1	11.7	8.6	(6)	
	DBStd	4.92	2.45	2.70	2.87	3.05	2.79	2.88	2.27	2.15	2.73	2.82	2.93	2.80	(7)	
	HDD10.0	177	54	28	14	5	0	0	0	0	0	0	16	61	(8)	
	HDD18.3	1612	297	229	202	152	83	27	9	8	22	80	200	302	(9)	
	CDD10.0	1822	15	32	70	106	187	264	301	303	271	191	66	18	(10)	
	CDD18.3	214	0	0	1	3	11	41	52	53	42	12	1	0	(11)	
	CDH23.3	4048	2	6	38	119	318	789	813	804	778	356	24	1	(12)	
	CDH26.7	1705	0	1	6	33	113	352	345	355	366	132	3	0	(13)	
(14) Wind	WSAvg	2.1	1.5	2.0	2.2	2.5	2.6	2.5	2.4	2.2	1.9	1.8	1.5	1.6	(14)	
(15) Precipitation	PrecAvg	1154	248	214	137	77	49	10	0	2	2	55	107	258	(15)	
	PrecMax	1825	761	643	478	236	176	36	3	30	10	153	282	586	(16)	
	PrecMin	629	24	10	6	16	1	1	0	0	0	1	8	60	(17)	
	PrecStd	346	175	144	134	58	53	11	1	7	3	39	79	166	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	24.2	27.3	30.6	33.4	36.8	37.0	36.4	36.8	33.9	26.2	19.9	(19)	
		MCWB	11.4	14.9	16.5	17.7	18.8	20.1	21.2	19.9	19.3	16.9	15.9	11.4	(20)	
	2%	DB	17.8	20.7	23.8	26.8	29.8	33.6	33.1	33.4	34.1	30.2	23.2	17.2	(21)	
		MCWB	11.4	13.4	14.9	16.4	17.2	19.1	19.7	19.2	18.5	16.1	14.2	11.2	(22)	
	5%	DB	15.6	18.2	21.2	23.6	26.9	30.8	30.4	30.7	31.2	27.4	20.4	15.4	(23)	
		MCWB	11.0	11.7	13.4	14.8	16.4	18.3	19.0	18.7	17.7	15.0	13.2	11.7	(24)	
	10%	DB	13.8	15.8	18.6	20.7	24.1	27.9	27.9	27.9	27.9	24.4	17.8	14.0	(25)	
		MCWB	10.8	11.0	12.3	13.5	15.1	17.3	18.2	18.1	17.1	14.5	12.4	11.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.7	17.5	19.0	20.2	21.7	21.8	21.4	21.0	18.9	17.4	15.1	(27)	
		MCDB	16.7	21.8	25.6	27.9	30.8	34.3	34.7	33.6	33.0	28.0	24.2	16.3	(28)	
	2%	WB	13.2	14.3	15.6	16.9	18.4	20.1	20.2	20.0	19.5	17.5	15.7	14.0	(29)	
		MCDB	15.2	18.8	22.6	25.4	28.1	31.7	31.7	31.4	31.2	27.2	20.3	15.1	(30)	
	5%	WB	12.3	13.1	14.3	15.4	16.9	18.8	19.2	19.0	18.4	16.4	14.6	12.9	(31)	
		MCDB	14.0	16.4	19.5	22.4	25.4	29.3	29.6	29.5	29.0	24.0	18.0	14.2	(32)	
	10%	WB	11.5	12.1	13.0	14.0	15.6	17.7	18.3	18.1	17.5	15.4	13.5	11.9	(33)	
		MCDB	13.1	14.6	16.7	19.6	23.0	27.0	27.7	27.6	27.1	22.1	16.5	13.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.5	12.5	13.4	14.8	16.5	16.6	16.7	17.6	16.2	12.9	10.7	(35)	
		MCDBR	15.1	16.6	17.9	19.0	20.5	21.9	21.6	22.2	23.8	22.3	17.6	13.7	(36)	
	5% WB	MCWBR	9.4	9.6	9.8	9.8	9.7	9.4	9.5	9.3	9.7	9.6	9.5	8.7	(37)	
		MCWBR	10.2	13.2	15.8	17.1	18.5	20.0	19.7	19.6	20.3	18.1	13.3	8.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.319	0.323	0.330	0.349	0.350	0.349	0.352	0.335	0.334	0.320	0.315	0.320	(40)	
		taud	2.556	2.545	2.528	2.437	2.468	2.477	2.468	2.515	2.509	2.591	2.578	2.553	(41)	
	Ebn at Noon		865	909	934	929	929	926	920	931	916	900	863	836	(42)	
		Edh at Noon	75	86	96	111	110	109	109	102	96	80	72	70	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.26	3.20	4.53	6.01	7.15	7.90	7.58	6.69	5.60	4.01	2.69	1.95	(44)	
	RadStd		0.36	0.44	0.46	0.49	0.45	0.36	0.29	0.25	0.23	0.29	0.20	0.33	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	N/A	(46)
(47)	Regional (39 neighbors)	N/A	N/A	-1.19	N/A	N/A	N/A	N/A	-104	+138	N/A	(47)

Nomenclature: See separate page